

HW6 - Computer Simulation II

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1

Generalizing the wolff algorithm to the XY model we can measure the mean energy per site, specific heat, and susceptibility for: $\beta \in [0, 2]$ in steps of 0.05 for 8×8 and 24×24 lattices

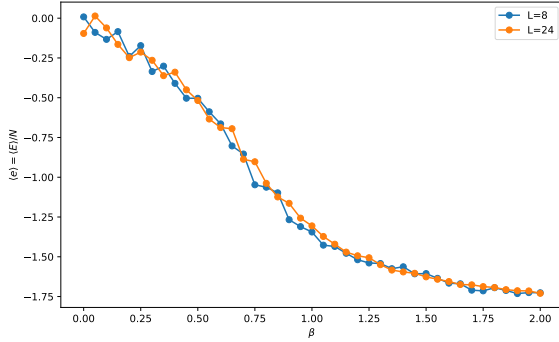


Figure 1: Mean energy per site $\times \beta$.

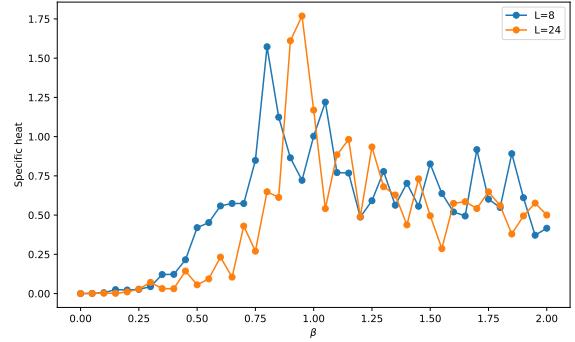


Figure 2: Specific heat $\times \beta$.

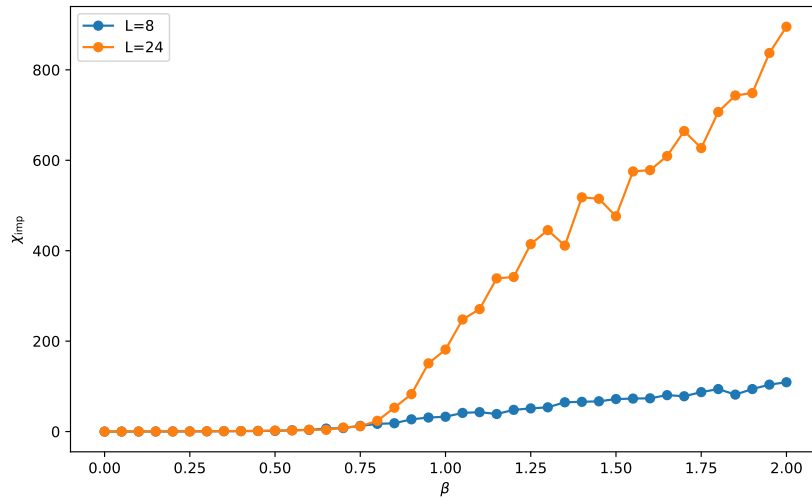


Figure 3: susceptibility $\times \beta$.

We can see the energy per site decreases with β (as expected) for both lattice sizes, the specific heat peaks around $\beta = 1$, and the susceptibility increases with β , showing a more pronounced increase around $\beta = 1$ for the larger lattice, indicating critical behavior.

2

For the 24×24 lattice, we compared the standard and improved susceptibility estimators at $\beta = 1.0, 0.9, 0.8, 0.7, 0.6$, and 0.5 .

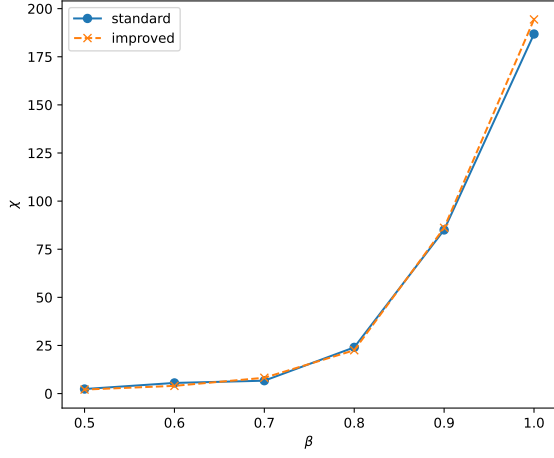


Figure 4: Standard x improved susceptibility.

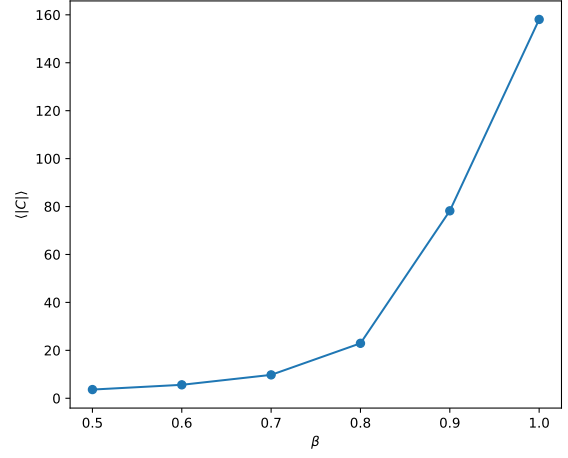


Figure 5: Mean cluster size at the comparison points.

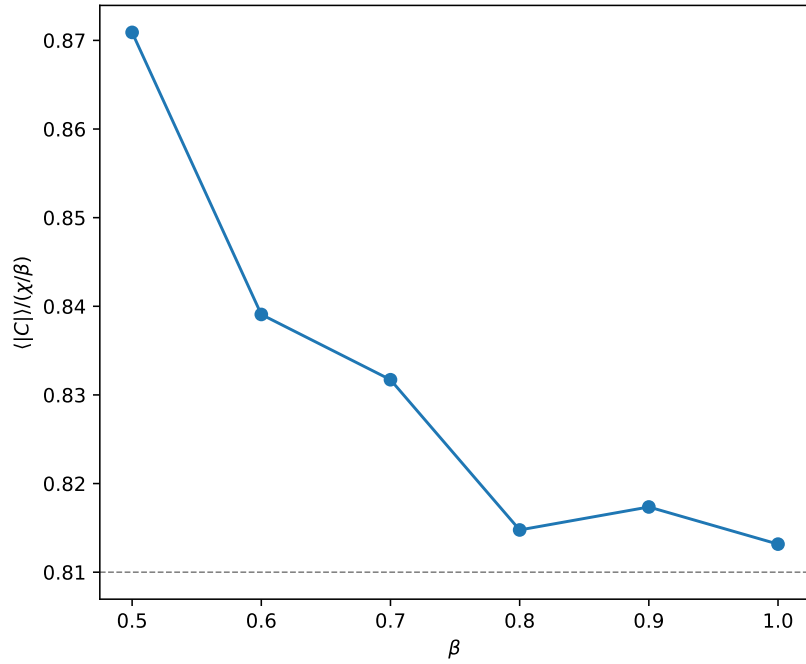


Figure 6: Ratio of mean cluster size to the improved susceptibility.

The improved susceptibility estimator remains close to the standard estimator being only slightly higher for bigger β values, the mean cluster size increases rapidly with β , and the ratio between the mean cluster size and the susceptibility remains above the expected 0.81, for small β values ranging from 0.87 to 0.82 only really approaching the expected value for smaller β , due to the bigger correlation lengths.